Solid mechanics for IBFE

Kinematics, strain, stress, and hyperelasticity

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IBFE Tutorials

$$\chi \longrightarrow \boldsymbol{F}_{\chi} \longrightarrow W \longrightarrow \boldsymbol{P}$$

motion, deformation, energy, stress

AI tools were used to assist with formatting, organization, and clarification of the instructor's original notes.

Roadmap

1. Notation and tensor bookkeeping
2. Configurations and the deformation map
3. Finite strain measures
4. Stress measures and transformations
5. Hyperelasticity and the IBFE force pipeline
6. Worked axial-deformation example
7. Neo-Hookean stress derivation and checks

$$\chi \longrightarrow \boldsymbol{F}_{\chi}, J \longrightarrow \boldsymbol{C}, \boldsymbol{E} \longrightarrow W \longrightarrow \boldsymbol{P} \longrightarrow \boldsymbol{F} \longrightarrow \boldsymbol{f}.$$

Module 0: purpose and scope

Goal

Build the minimum solid-mechanics toolkit needed to follow IBFE derivations.

Audience

Advanced undergraduates and graduate students who are new to continuum mechanics.

This module establishes:

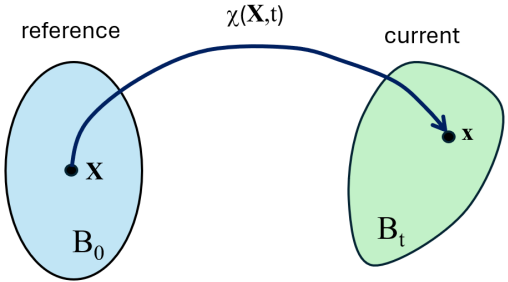
- reference and current coordinates,
- tensor types and indices,
- material and spatial derivatives,
- distinct symbols for deformation and force.

Configurations, coordinates, and fields

- **Reference configuration**: $B_0 \subset \mathbb{R}^d$, with material coordinate $\boldsymbol{X}$.
- **Current configuration**: $B_t \subset \mathbb{R}^d$, with spatial coordinate $\boldsymbol{x}$.
- **Motion**: $\chi(\cdot, t) : B_0 \to B_t$, with

$$\boldsymbol{x} = \chi(\boldsymbol{X}, t).$$

Material fields use $\boldsymbol{X}$ as their independent coordinate. Spatial fields use $\boldsymbol{x}$.



A material point keeps its label $\boldsymbol{X}$ while its current position changes.

Scalars, vectors, and second-order tensors

Object	Components	Example
Scalar	a	energy density W
Vector	v_i or v_I	velocity, normal, force
Spatial tensor	A_{ij}	Cauchy stress σ_{ij}
Material tensor	A_{IJ}	Green strain E_{IJ}
Two-point tensor	A_{iI}	$(F_\chi)_{iI}$, P_{iI}

A two-point tensor maps a material direction into a spatial direction. Both the deformation gradient and first Piola stress have one spatial and one material index.

$$i, j, k: \text{ spatial}, \qquad I, J, K: \text{ material}.$$

Repeated indices are summed unless stated otherwise.

Material and spatial fields

Reference domain B_0	Current domain B_t or fluid domain Ω
$\boldsymbol{X}$: material coordinate	$\boldsymbol{x}$: spatial coordinate
$\chi(\boldsymbol{X}, t)$: motion	$\boldsymbol{u}(\boldsymbol{x}, t)$: velocity
$\boldsymbol{F}_\chi = \nabla_X \chi$: deformation gradient	$\boldsymbol{\sigma}(\boldsymbol{x}, t)$: Cauchy stress
$J = \det \boldsymbol{F}_\chi$: volume ratio	$p(\boldsymbol{x}, t)$: pressure
$\boldsymbol{P}(\boldsymbol{X}, t)$: first Piola stress	$\boldsymbol{f}(\boldsymbol{x}, t)$: Eulerian force density
$\boldsymbol{F}(\boldsymbol{X}, t)$: Lagrangian force density	

Always check the independent coordinate before differentiating or integrating a field.

Material and spatial derivatives

Material gradient

$$\nabla_X(\cdot)$$

differentiates with respect to $\boldsymbol{X}$. For the motion,

$$(\nabla_X \chi)_{iI} = \frac{\partial \chi_i}{\partial X_I}.$$

Spatial gradient

$$\nabla_x(\cdot)$$

differentiates with respect to $\boldsymbol{x}$. For velocity,

$$(\nabla_x \boldsymbol{u})_{ij} = \frac{\partial u_i}{\partial x_j}.$$

IBFE structural kinematics and constitutive stress are normally evaluated on B_0, so ∇_X appears throughout the finite-element formulation.

Deformation gradient and Lagrangian force

IB and IBFE papers often use bold F for a Lagrangian force density, while solid-mechanics texts use bold F for the deformation gradient.

Kinematics

$$\boldsymbol{F}_\chi(\boldsymbol{X}, t) := \nabla_{\boldsymbol{X}} \chi(\boldsymbol{X}, t)$$

is a two-point tensor.

Forcing

$$\boldsymbol{F}(\boldsymbol{X}, t)$$

is a Lagrangian vector force density.

$$\boldsymbol{F}_\chi \neq \boldsymbol{F}.$$

The subscript on $\boldsymbol{F}_\chi$ keeps kinematics distinct from forcing.

$$\chi(\cdot, t) : B_0 \to B_t, \qquad \boldsymbol{x} = \chi(\boldsymbol{X}, t).$$

The displacement is

$$\boldsymbol{d}(\boldsymbol{X}, t) = \chi(\boldsymbol{X}, t) - \boldsymbol{X}.$$

This module develops two local measures from the motion:

$$\boldsymbol{F}_\chi = \nabla_X \chi, \qquad J = \det \boldsymbol{F}_\chi.$$

$\boldsymbol{F}_\chi$ maps line elements and J maps volume elements.

Nearby material points under the motion

Choose neighboring material points

$$\boldsymbol{X}, \qquad \boldsymbol{X} + d\boldsymbol{X}.$$

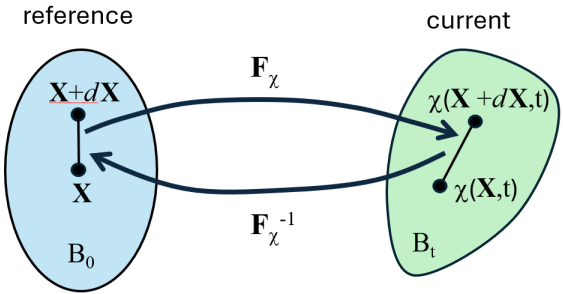
Their current positions are

$$\boldsymbol{x} = \chi(\boldsymbol{X}, t), \qquad \boldsymbol{x} + d\boldsymbol{x} = \chi(\boldsymbol{X} + d\boldsymbol{X}, t).$$

Thus

$$d\boldsymbol{x} = \chi(\boldsymbol{X} + d\boldsymbol{X}, t) - \chi(\boldsymbol{X}, t).$$

The deformation gradient is the linear map that approximates this change for small $d\boldsymbol{X}$.



Deformation-gradient definition

Linearizing the motion about $\boldsymbol{X}$ gives

$$d\boldsymbol{x} = \boldsymbol{F}_\chi(\boldsymbol{X}, t)\, d\boldsymbol{X} + o(\|d\boldsymbol{X}\|),$$

with

$$\boxed{\boldsymbol{F}_\chi(\boldsymbol{X}, t) = \nabla_X \chi(\boldsymbol{X}, t)}.$$

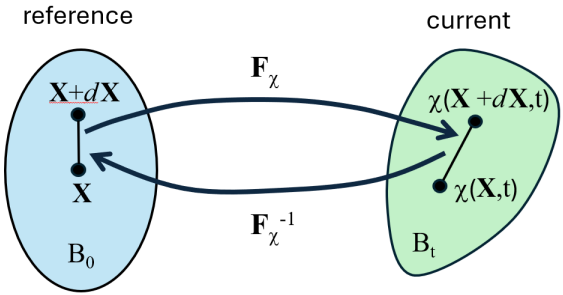
In components,

$$(F_\chi)_{iI} = \frac{\partial \chi_i}{\partial X_I}.$$

Columns of $\boldsymbol{F}_\chi$
Images of the reference basis vectors.

Information in $\boldsymbol{F}_\chi$
Stretch, shear, and rotation near one material point.

Inverse differential map at matching points



If $\chi(\cdot, t)$ is locally invertible and $\boldsymbol{x} = \chi(\boldsymbol{X}, t)$, then

$$d\boldsymbol{X} = \boldsymbol{F}_\chi(\boldsymbol{X}, t)^{-1} d\boldsymbol{x} + o(\|d\boldsymbol{x}\|).$$

The spatial gradient of the inverse motion is

$$\boxed{\nabla_x \chi^{-1}(\boldsymbol{x}, t) = \boldsymbol{F}_\chi(\chi^{-1}(\boldsymbol{x}, t), t)^{-1}}.$$

The composition with χ^{-1} ensures that both tensors refer to the same material point.

Jacobian and local volume change

$$\boxed{J(\boldsymbol{X}, t) = \det \boldsymbol{F}_\chi(\boldsymbol{X}, t)}, \qquad dv = J\, dV.$$

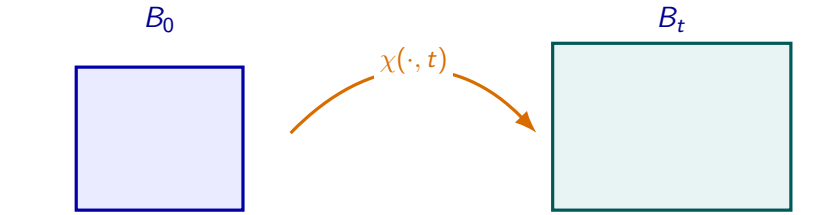
- $J > 1$: local expansion
- $0 < J < 1$: local compression
- $J = 1$: local volume preservation

An admissible deformation normally requires

$$J > 0.$$

$J = 0$ collapses material volume. $J < 0$ reverses orientation and indicates element inversion.

Visualizing the Jacobian as a local measure ratio



$$dv = J\,dV, \qquad J = \det \boldsymbol{F}_\chi$$

In two dimensions the corresponding area relation is $da = J\,dA$.

Jacobian as the product of principal stretches

For a diagonal deformation in d dimensions,

$$\boldsymbol{F}_\chi = \operatorname{diag}(\lambda_1, \ldots, \lambda_d), \qquad J = \prod_{k=1}^{d} \lambda_k.$$

Two dimensions

$$da = J\, dA = \lambda_1 \lambda_2\, dA.$$

Three dimensions

$$dv = J\, dV = \lambda_1 \lambda_2 \lambda_3\, dV.$$

Volume preservation does not imply zero strain. A simple shear has $J = 1$ and still changes angles and lengths.

Micro-check: deformation gradients and Jacobians

A. Uniform stretch

$$\chi(\boldsymbol{X}) = (\lambda X_1, \lambda X_2, \lambda X_3).$$

1. Compute $\boldsymbol{F}_\chi = \nabla_X \chi$.
2. Compute $J = \det \boldsymbol{F}_\chi$.
3. Interpret the volume change.

B. Simple shear

$$\chi(\boldsymbol{X}) = (X_1 + \gamma X_2, X_2, X_3).$$

1. Compute each $(F_\chi)_{iI}$.
2. Compute J.
3. Decide whether the map is strain free.

Micro-check solution: uniform stretch

From

$$\chi_1 = \lambda X_1, \qquad \chi_2 = \lambda X_2, \qquad \chi_3 = \lambda X_3,$$

we obtain

$$\boldsymbol{F}_\chi = \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \lambda \boldsymbol{I}.$$

Therefore

$$\boxed{J = \det \boldsymbol{F}_\chi = \lambda^3}.$$

Every length scales by λ, every area by λ^2, and every volume by λ^3.

Micro-check solution: simple shear

For

$$\chi(\boldsymbol{X}) = (X_1 + \gamma X_2, X_2, X_3),$$

$$\boldsymbol{F}_\chi = \begin{bmatrix} 1 & \gamma & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Because this matrix is upper triangular,

$$\boxed{J = 1}.$$

Simple shear preserves volume but changes line lengths and angles. The strain measures in Module 2 detect those changes.

The displacement gradient is not a finite strain

Since

$$\chi(\boldsymbol{X}, t) = \boldsymbol{X} + \boldsymbol{d}(\boldsymbol{X}, t),$$

we have

$$\nabla_X \boldsymbol{d} = \boldsymbol{F}_\chi - \boldsymbol{I}.$$

What it is

A first derivative of displacement that contains stretch, shear, and rotation.

Small-strain approximation

$$\varepsilon = \frac{1}{2}\left(\nabla_X \boldsymbol{d} + (\nabla_X \boldsymbol{d})^T\right).$$

Finite deformation requires nonlinear strain measures that remain unchanged under rigid-body motion.

Module 1 wrap-up

- The motion maps material points: $\boldsymbol{x} = \chi(\boldsymbol{X}, t)$.
- The deformation gradient maps line elements:

$$\boldsymbol{F}_\chi = \nabla_X \chi, \qquad d\boldsymbol{x} \approx \boldsymbol{F}_\chi \, d\boldsymbol{X}.$$

- The Jacobian maps volume elements:

$$J = \det \boldsymbol{F}_\chi, \qquad dv = J \, dV.$$

- The material and spatial gradients act on different independent coordinates.

Quick check: if $\chi(\boldsymbol{X}) = \lambda \boldsymbol{X}$ in three dimensions, then $\boldsymbol{F}_\chi = \lambda \boldsymbol{I}$ and $J = \lambda^3$.

Module 2: requirements for a finite strain

A useful finite-strain measure must:

- detect stretch and shear,
- ignore rigid translation and rigid rotation,
- remain meaningful for large deformation,
- reduce to linearized strain when $\|\nabla_X \boldsymbol{d}\| \ll 1$.

The deformation gradient contains both deformation and rotation. Strain tensors built from $\boldsymbol{F}_\chi^T \boldsymbol{F}_\chi$ or $\boldsymbol{F}_\chi \boldsymbol{F}_\chi^T$ separate those effects.

Rotation and stretch should feel different

Pure rotation

$$\boldsymbol{F}_\chi = \boldsymbol{Q}, \qquad \boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{I}.$$

Lengths and angles remain unchanged.

desired strain $= \boldsymbol{0}$.

Pure stretch

$$\boldsymbol{F}_\chi = \text{diag}(\lambda_1, \lambda_2, \lambda_3).$$

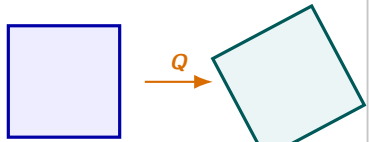
At least one length changes when $\lambda_i \neq 1$.

desired strain $\neq \boldsymbol{0}$.

This test immediately rules out $\boldsymbol{F}_\chi - \boldsymbol{I}$: a finite rotation gives $\boldsymbol{Q} - \boldsymbol{I} \neq \boldsymbol{0}$.

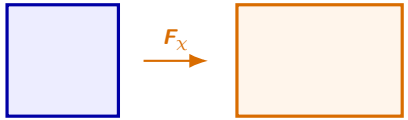
Visual check: rigid rotation versus genuine stretch

Pure rotation



$\boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{I} \quad \Rightarrow \quad \text{strain} = \boldsymbol{0}$

Pure stretch



$\lambda_i \neq 1 \quad \Rightarrow \quad \text{strain} \neq \boldsymbol{0}$

Displacement gradient and linearized strain

With

$$\chi = \boldsymbol{X} + \boldsymbol{d}, \qquad \boldsymbol{F}_\chi = \boldsymbol{I} + \nabla_X \boldsymbol{d},$$

the linearized strain is

$$\boxed{\boldsymbol{\varepsilon} = \mathrm{sym}(\nabla_X \boldsymbol{d}) = \frac{1}{2}\left(\nabla_X \boldsymbol{d} + (\nabla_X \boldsymbol{d})^T\right).}$$

This approximation assumes small displacement gradients, including small rotations. Finite rotations require a nonlinear strain measure.

Right and left Cauchy–Green tensors

$$\boldsymbol{C} = \boldsymbol{F}_\chi^T \boldsymbol{F}_\chi \qquad \text{(material)}$$

$$\boldsymbol{B} = \boldsymbol{F}_\chi \boldsymbol{F}_\chi^T \qquad \text{(spatial)}$$

When $J > 0$, both are symmetric positive definite and share the eigenvalues λ_i^2.

Under a superposed spatial rotation $\boldsymbol{F}_\chi^\star = \boldsymbol{Q}\boldsymbol{F}_\chi$,

$$\boldsymbol{C}^\star = \boldsymbol{C}, \qquad \boldsymbol{B}^\star = \boldsymbol{Q}\boldsymbol{B}\boldsymbol{Q}^T.$$

$\boldsymbol{C}$ remains unchanged. $\boldsymbol{B}$ rotates with the spatial frame, while its scalar invariants remain unchanged.

Green–Lagrange strain

$$\boxed{\boldsymbol{E} = \frac{1}{2}(\boldsymbol{C} - \boldsymbol{I}) = \frac{1}{2}(\boldsymbol{F}_\chi^T \boldsymbol{F}_\chi - \boldsymbol{I})}.$$

In components,

$$E_{IJ} = \frac{1}{2}(C_{IJ} - \delta_{IJ}), \qquad C_{IJ} = \sum_{i=1}^{d} (F_\chi)_{iI}(F_\chi)_{iJ}.$$

Configuration
$\boldsymbol{E}$ acts on material line elements.

Small-deformation limit
$\boldsymbol{E} = \varepsilon + O(\|\nabla_X \boldsymbol{d}\|^2)$.

Green–Lagrange strain and length change

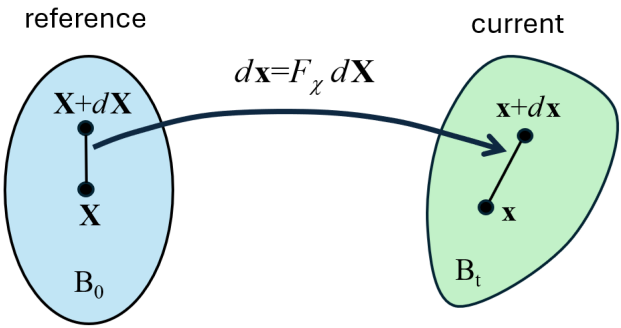
For a material line element,

$$d\boldsymbol{x} = \boldsymbol{F}_\chi \, d\boldsymbol{X}.$$

Then

$$\|d\boldsymbol{x}\|^2 - \|d\boldsymbol{X}\|^2 = 2\, d\boldsymbol{X} \cdot \boldsymbol{E}\, d\boldsymbol{X}.$$

$\boldsymbol{E} = \boldsymbol{0}$ means that local lengths are preserved. For a compatible deformation of a connected body, this corresponds to rigid motion.



$$|d\boldsymbol{x}|^2 - |d\boldsymbol{X}|^2 \;=\; 2\, d\boldsymbol{X} \cdot \boldsymbol{E}\, d\boldsymbol{X}$$

The strain is expressed using the original material line element.

Euler–Almansi strain

The spatial finite-strain measure is

$$\boxed{\boldsymbol{e} = \frac{1}{2}(\boldsymbol{I} - \boldsymbol{B}^{-1}) = \frac{1}{2}\left[\boldsymbol{I} - (\boldsymbol{F}_\chi \boldsymbol{F}_\chi^T)^{-1}\right]}.$$

Green–Lagrange
$\boldsymbol{E}$ acts on vectors in the reference configuration.

Euler–Almansi
$\boldsymbol{e}$ acts on vectors in the current configuration.

IBFE derivations usually favor $\boldsymbol{F}_\chi$, $\boldsymbol{C}$, and $\boldsymbol{E}$ because finite-element integration occurs on B_0.

Euler–Almansi strain and current line elements

Take a current line element $d\boldsymbol{x}$. Its material counterpart is

$$d\boldsymbol{X} = \boldsymbol{F}_\chi^{-1} d\boldsymbol{x}.$$

Therefore

$$\|d\boldsymbol{x}\|^2 - \|d\boldsymbol{X}\|^2 = d\boldsymbol{x} \cdot (\boldsymbol{I} - \boldsymbol{B}^{-1}) d\boldsymbol{x} = 2\, d\boldsymbol{x} \cdot \boldsymbol{e}\, d\boldsymbol{x}.$$

The two finite-strain measures are related by

$$\boxed{\boldsymbol{e} = \boldsymbol{F}_\chi^{-T} \boldsymbol{E} \boldsymbol{F}_\chi^{-1}}.$$

The formulas describe the same deformation using different coordinates.

Micro-check: pure rotation

Let

$$\chi(\boldsymbol{X}) = \boldsymbol{Q}\boldsymbol{X} + \boldsymbol{c}, \qquad \boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{I}, \qquad \det \boldsymbol{Q} = 1.$$

Then

$$\boldsymbol{F}_\chi = \boldsymbol{Q}, \qquad \boldsymbol{C} = \boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{I}, \qquad \boldsymbol{E} = \boldsymbol{0}.$$

Also,

$$\boldsymbol{B} = \boldsymbol{Q}\boldsymbol{Q}^T = \boldsymbol{I}, \qquad \boldsymbol{e} = \boldsymbol{0}.$$

Both finite-strain measures pass the rigid-rotation test.

Micro-check: simple shear

For

$$\boldsymbol{F}_\chi = \begin{bmatrix} 1 & \gamma & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

compute

$$\boldsymbol{C} = \begin{bmatrix} 1 & \gamma & 0 \\ \gamma & 1+\gamma^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \boldsymbol{E} = \begin{bmatrix} 0 & \gamma/2 & 0 \\ \gamma/2 & \gamma^2/2 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Linearized strain retains the terms proportional to γ. Finite strain also contains the normal-strain term $\gamma^2/2$.

Polar decomposition: $\boldsymbol{F}_\chi = \boldsymbol{RU} = \boldsymbol{VR}$

Every deformation gradient with $\det \boldsymbol{F}_\chi > 0$ factors **uniquely** as

$$\boxed{\boldsymbol{F}_\chi = \boldsymbol{RU} = \boldsymbol{VR}}, \qquad \boldsymbol{R}^T\boldsymbol{R} = \boldsymbol{I}, \qquad \det \boldsymbol{R} = 1,$$

$$\boldsymbol{U} = \sqrt{\boldsymbol{F}_\chi^T \boldsymbol{F}_\chi} = \sqrt{\boldsymbol{C}}, \qquad \boldsymbol{V} = \sqrt{\boldsymbol{F}_\chi \boldsymbol{F}_\chi^T} = \sqrt{\boldsymbol{B}}, \qquad \boldsymbol{V} = \boldsymbol{RUR}^T.$$

The right stretch $\boldsymbol{U}$ and the left stretch $\boldsymbol{V}$ are symmetric positive definite; $\boldsymbol{R}$ is proper orthogonal.

Reading *RU*
Stretch first along the eigenvectors of $\boldsymbol{U}$ by the principal stretches λ_i, then rotate by $\boldsymbol{R}$.

Reading *VR*
Rotate first by the same $\boldsymbol{R}$, then stretch by the same λ_i in the rotated frame.

The eigenvalues of $\boldsymbol{U}$ are the principal stretches λ_i; the eigenvalues of $\boldsymbol{C} = \boldsymbol{U}^2$ are λ_i^2.

Consequence. Objectivity demands $W(\boldsymbol{Q}\boldsymbol{F}_\chi) = W(\boldsymbol{F}_\chi)$ for every rotation $\boldsymbol{Q}$. Take $\boldsymbol{Q} = \boldsymbol{R}^T$: $W(\boldsymbol{F}_\chi) = W(\boldsymbol{R}^T\boldsymbol{RU}) = W(\boldsymbol{U}) =: \widehat{W}(\boldsymbol{U}) = \widetilde{W}(\boldsymbol{C})$. A strain energy can see $\boldsymbol{F}_\chi$ only through $\boldsymbol{U}$, equivalently through $\boldsymbol{C} = \boldsymbol{U}^2$.

Polar decomposition: worked check

Take the simple shear of the previous slide in the X_1X_2 plane with $\gamma = 1$:

$$\boldsymbol{F}_\chi = \begin{bmatrix} 1 & \gamma \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad \boldsymbol{C} = \boldsymbol{F}_\chi^T \boldsymbol{F}_\chi = \begin{bmatrix} 1 & \gamma \\ \gamma & 1+\gamma^2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Principal stretches. $\lambda_{1,2} = \frac{1}{2}\left(\sqrt{4+\gamma^2} \pm \gamma\right)$, so $\lambda_1 = \frac{\sqrt{5}+1}{2} \approx 1.618$, $\lambda_2 = \frac{\sqrt{5}-1}{2} \approx 0.618$, and $\lambda_1\lambda_2 = 1 = \det \boldsymbol{F}_\chi$. The eigenvalues of $\boldsymbol{C}$ are $\lambda_1^2 \approx 2.618$ and $\lambda_2^2 \approx 0.382$.

$$\boldsymbol{U} = \sqrt{\boldsymbol{C}} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \approx \begin{bmatrix} 0.894 & 0.447 \\ 0.447 & 1.342 \end{bmatrix}, \qquad \boldsymbol{R} = \boldsymbol{F}_\chi \boldsymbol{U}^{-1} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \approx \begin{bmatrix} 0.894 & 0.447 \\ -0.447 & 0.894 \end{bmatrix}.$$

Checks: $\boldsymbol{U}^2 = \boldsymbol{C}$, $\boldsymbol{R}^T\boldsymbol{R} = \boldsymbol{I}$, $\det \boldsymbol{R} = 1$, and $\boldsymbol{R}$ is a clockwise rotation through θ with $\tan\theta = \gamma/2 = \frac{1}{2}$, that is $\theta \approx 26.6^\circ$.

Simple shear is a stretch of λ_1 along one principal direction, a compression $\lambda_2 = 1/\lambda_1$ along the other, plus a rigid rotation by θ; that rotation is exactly what $\boldsymbol{C}$ and $\boldsymbol{E}$ discard.

Strain invariants and volume change

In three dimensions,

$$I_1 = \operatorname{tr} \boldsymbol{C}, \quad I_2 = \frac{1}{2}\left[(\operatorname{tr} \boldsymbol{C})^2 - \operatorname{tr}(\boldsymbol{C}^2)\right], \quad I_3 = \det \boldsymbol{C} = J^2.$$

Raw invariants

I_1 and I_2 respond to both shape change and volume change. I_3 measures volume change directly.

Isochoric invariants

Using $\bar{\boldsymbol{F}}_\chi = J^{-1/3}\boldsymbol{F}_\chi$ and $\bar{\boldsymbol{C}} = \bar{\boldsymbol{F}}_\chi{}^T \bar{\boldsymbol{F}}_\chi$,

$$\bar{I}_1 = \operatorname{tr} \bar{\boldsymbol{C}}, \qquad \bar{I}_2 = \tfrac{1}{2}[(\operatorname{tr} \bar{\boldsymbol{C}})^2 - \operatorname{tr}(\bar{\boldsymbol{C}}^2)].$$

Module 2 wrap-up

- $\nabla_X \boldsymbol{d} = \boldsymbol{F}_\chi - \boldsymbol{I}$ is the displacement gradient.
- $\boldsymbol{C} = \boldsymbol{F}_\chi^T \boldsymbol{F}_\chi$ and $\boldsymbol{B} = \boldsymbol{F}_\chi \boldsymbol{F}_\chi^T$ encode stretch.
- $\boldsymbol{E} = \frac{1}{2}(\boldsymbol{C} - \boldsymbol{I})$ is a material finite-strain measure.
- $\boldsymbol{e} = \frac{1}{2}(\boldsymbol{I} - \boldsymbol{B}^{-1})$ is its spatial counterpart.
- $I_3 = J^2$. Modified invariants isolate isochoric deformation.

Quick check: $J = 1$ implies $I_3 = 1$, but it does not imply $\boldsymbol{E} = \boldsymbol{0}$.

Fluid mechanics normally uses the Cauchy stress

$$\boldsymbol{\sigma}(\boldsymbol{x}, t)$$

in the current configuration. IBFE constitutive calculations usually use reference coordinates and the stresses

$$\boldsymbol{P}(\boldsymbol{X}, t) \qquad \text{and} \qquad \boldsymbol{S}(\boldsymbol{X}, t).$$

Each stress measure pairs with a particular normal, area measure, or deformation rate. Conversions between them describe the same physical force or power.

Cauchy stress and current traction

For a current surface with outward unit normal $\boldsymbol{n}$,

$$\boxed{\boldsymbol{t}(\boldsymbol{n}) = \boldsymbol{\sigma}\,\boldsymbol{n}}.$$

The differential force acting on current area da is

$$d\boldsymbol{f} = \boldsymbol{\sigma}\,\boldsymbol{n}\,da.$$

In components,

$$t_i = \sigma_{ij} n_j.$$

The Cauchy stress maps a current normal to force per current area. Its diagonal entries describe normal stress and its off-diagonal entries describe shear stress in the chosen basis.

Traction vectors on coordinate planes

$$\boldsymbol{t}(\boldsymbol{n}) = \boldsymbol{\sigma}\boldsymbol{n}.$$

Choosing each coordinate normal selects one column of the Cauchy-stress matrix.

For example,

$$\boldsymbol{t}(\boldsymbol{e}_1) = \begin{bmatrix} \sigma_{11} \\ \sigma_{21} \\ \sigma_{31} \end{bmatrix}.$$

$$\mathbf{t}^{(\mathbf{e}_j)} = \boldsymbol{\sigma}\mathbf{e}_j, \qquad \begin{aligned} \mathbf{t}^{(\mathbf{e}_1)} &= \sigma_{11}\mathbf{e}_1 + \sigma_{21}\mathbf{e}_2 + \sigma_{31}\mathbf{e}_3 \\ \mathbf{t}^{(\mathbf{e}_2)} &= \sigma_{12}\mathbf{e}_1 + \sigma_{22}\mathbf{e}_2 + \sigma_{32}\mathbf{e}_3 \\ \mathbf{t}^{(\mathbf{e}_3)} &= \sigma_{13}\mathbf{e}_1 + \sigma_{23}\mathbf{e}_2 + \sigma_{33}\mathbf{e}_3 \end{aligned} \qquad (3.3.7)$$

Eqns. 3.3.6-7 are illustrated in Fig. 3.3.2.

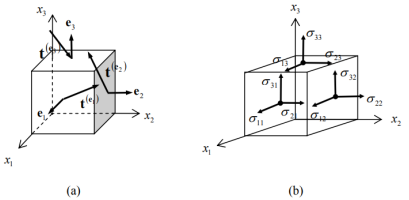


Figure 3.3.2: traction acting on surfaces with normals in the coordinate directions; (a) traction vectors, (b) stress components

Source: P. A. Kelly, *Solid Mechanics Part III: Foundations of Continuum Solid Mechanics*, §3.3, Fig. 3.3.2 (University of Auckland lecture notes, https://pkel015.connect.amazon.auckland.ac.nz/SolidMechanicsBooks/). The figure's equation numbers refer to those notes.

First Piola–Kirchhoff stress

The first Piola stress is a two-point tensor defined on B_0:

$$\boldsymbol{P}(\boldsymbol{X}, t) \in \mathbb{R}^{d \times d}.$$

For a reference surface with outward unit normal $\boldsymbol{N}$ and reference area dA,

$$\boxed{d\boldsymbol{f} = \boldsymbol{P}\boldsymbol{N}\, dA}.$$

Thus

$$\boldsymbol{t}_0(\boldsymbol{N}) = \boldsymbol{P}\boldsymbol{N}$$

is force per reference area, directed in the current configuration.

$\boldsymbol{P}$ is generally nonsymmetric because its two indices refer to different configurations.

Matching reference and current surface patches

The same physical force can be described on corresponding patches:

$$d\boldsymbol{f} = \boldsymbol{\sigma n}\, da = \boldsymbol{PN}\, dA.$$

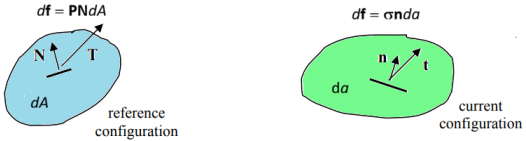


Figure 3.5.1: Traction vectors

- $\boldsymbol{\sigma n}$: force per current area
- $\boldsymbol{PN}$: force per reference area

The area vectors differ. Nanson's formula supplies the missing geometric transformation.

Source: Kelly, *Solid Mechanics Part III*, §3.5, Fig. 3.5.1 (same notes). NB: the figure labels the reference traction $\boldsymbol{PN}$ as $\boldsymbol{T}$; in this course that quantity is $\boldsymbol{t}_0$, and $\boldsymbol{T}$ is reserved for the transmission force $-\boldsymbol{PN}$.

Nanson's formula

Corresponding oriented area elements satisfy

$$\boxed{\boldsymbol{n}\, da = J\, \boldsymbol{F}_\chi^{-T}\, \boldsymbol{N}\, dA}.$$

Reference area vector

$$\boldsymbol{N}\, dA$$

uses the material normal and reference area.

Current area vector

$$\boldsymbol{n}\, da$$

uses the spatial normal and current area.

In three dimensions, the identity follows by transforming two tangent vectors and using

$$(\boldsymbol{F}_\chi \boldsymbol{a}) \times (\boldsymbol{F}_\chi \boldsymbol{b}) = J \boldsymbol{F}_\chi^{-T} (\boldsymbol{a} \times \boldsymbol{b}).$$

Piola transformation

Start from equality of force on matching patches:

$$\boldsymbol{\sigma}\boldsymbol{n}\, da = \boldsymbol{P}\boldsymbol{N}\, dA.$$

Insert Nanson's formula:

$$\boldsymbol{\sigma}\left(J\boldsymbol{F}_{\chi}^{-T}\boldsymbol{N}\, dA\right) = \boldsymbol{P}\boldsymbol{N}\, dA.$$

Because the relation holds for every $\boldsymbol{N}$,

$$\boxed{\boldsymbol{P} = J\,\boldsymbol{\sigma}\,\boldsymbol{F}_{\chi}^{-T}}.$$

Equivalently,

$$\boxed{\boldsymbol{\sigma} = J^{-1}\boldsymbol{P}\boldsymbol{F}_{\chi}^{T}}.$$

Second Piola–Kirchhoff stress

Pull the remaining spatial index of $\boldsymbol{P}$ back to the reference configuration:

$$\boxed{\boldsymbol{S} = \boldsymbol{F}_\chi^{-1}\boldsymbol{P}}, \qquad \boxed{\boldsymbol{P} = \boldsymbol{F}_\chi \boldsymbol{S}}.$$

Configuration

$\boldsymbol{S}$ has two material indices and acts entirely in the reference description.

Energy pairing

$$\boldsymbol{P} : \dot{\boldsymbol{F}}_\chi = \frac{1}{2}\boldsymbol{S} : \dot{\boldsymbol{C}}.$$

If the Cauchy stress is symmetric, then $\boldsymbol{S}$ is symmetric.

Dimensional, symmetry, and power checks

Measure	Area basis	Symmetry	Power pairing
$\boldsymbol{\sigma}$	current	symmetric*	$J\boldsymbol{\sigma} : \boldsymbol{D}$
$\boldsymbol{P}$	reference	generally no	$\boldsymbol{P} : \dot{\boldsymbol{F}}_\chi$
$\boldsymbol{S}$	reference	symmetric*	$\frac{1}{2}\boldsymbol{S} : \dot{\boldsymbol{C}}$

$$\boldsymbol{P} : \dot{\boldsymbol{F}}_\chi = \frac{1}{2}\boldsymbol{S} : \dot{\boldsymbol{C}} = J\boldsymbol{\sigma} : \boldsymbol{D}, \qquad \boldsymbol{D} = \operatorname{sym}(\nabla_x \boldsymbol{u}).$$

*Symmetry follows for a classical nonpolar continuum satisfying angular-momentum balance.

Module 3 wrap-up

- Current traction: $\boldsymbol{t} = \boldsymbol{\sigma}\boldsymbol{n}$.
- Reference traction: $\boldsymbol{t}_0 = \boldsymbol{P}\boldsymbol{N}$.
- Oriented areas:

$$\boldsymbol{n}\, da = J\boldsymbol{F}_\chi^{-T}\boldsymbol{N}\, dA.$$

- Stress conversions:

$$\boldsymbol{P} = J\boldsymbol{\sigma}\boldsymbol{F}_\chi^{-T}, \qquad \boldsymbol{\sigma} = J^{-1}\boldsymbol{P}\boldsymbol{F}_\chi^{T}, \qquad \boldsymbol{S} = \boldsymbol{F}_\chi^{-1}\boldsymbol{P}.$$

The normal, area measure, and stress tensor must belong to matching configurations.

IBFE notation bridge

Object	Symbol here	Role in IBFE
Deformation gradient	$\boldsymbol{F}_\chi = \nabla_X \chi$	structural kinematics
First Piola stress	$\boldsymbol{P}$	constitutive response
Lagrangian force	$\boldsymbol{F}$	weak structural force
Eulerian force	$\boldsymbol{f}$	fluid momentum source
Cauchy stress	$\boldsymbol{\sigma}$	current-configuration stress

$$W(\boldsymbol{F}_\chi) \longrightarrow \boldsymbol{P} \longrightarrow \boldsymbol{F} \longrightarrow \boldsymbol{f}.$$

A hyperelastic material stores energy according to a scalar density per reference volume:

$$W : \mathbb{R}^{d\times d} \to \mathbb{R}, \qquad \boldsymbol{F}_\chi \mapsto W(\boldsymbol{F}_\chi).$$

The first Piola stress follows by differentiation:

$$\boxed{\boldsymbol{P} = \frac{\partial W}{\partial \boldsymbol{F}_\chi}}.$$

One scalar function encodes the constitutive response. Differentiating once gives stress. Differentiating again gives the material tangent used in nonlinear solvers.

Derivative of a scalar with respect to a matrix

The derivative has the same tensor shape as its matrix argument:

$$\left(\frac{\partial W}{\partial \boldsymbol{F}_\chi}\right)_{iI} := \frac{\partial W}{\partial (F_\chi)_{iI}}.$$

In three dimensions,

$$\frac{\partial W}{\partial \boldsymbol{F}_\chi} = \begin{bmatrix} \dfrac{\partial W}{\partial (F_\chi)_{11}} & \dfrac{\partial W}{\partial (F_\chi)_{12}} & \dfrac{\partial W}{\partial (F_\chi)_{13}} \\ \dfrac{\partial W}{\partial (F_\chi)_{21}} & \dfrac{\partial W}{\partial (F_\chi)_{22}} & \dfrac{\partial W}{\partial (F_\chi)_{23}} \\ \dfrac{\partial W}{\partial (F_\chi)_{31}} & \dfrac{\partial W}{\partial (F_\chi)_{32}} & \dfrac{\partial W}{\partial (F_\chi)_{33}} \end{bmatrix}.$$

Shape check: W is scalar, while $\partial W / \partial \boldsymbol{F}_\chi \in \mathbb{R}^{d \times d}$.

Directional derivative and double contraction

Perturb $\boldsymbol{F}_\chi$ in an arbitrary direction $\delta\boldsymbol{F}_\chi$:

$$\delta W := \frac{d}{d\varepsilon} W(\boldsymbol{F}_\chi + \varepsilon\,\delta\boldsymbol{F}_\chi)\bigg|_{\varepsilon=0}.$$

The matrix derivative is defined by

$$\boxed{\delta W = \frac{\partial W}{\partial \boldsymbol{F}_\chi} : \delta\boldsymbol{F}_\chi = \boldsymbol{P} : \delta\boldsymbol{F}_\chi}.$$

The double contraction is the Frobenius inner product:

$$\boldsymbol{A} : \boldsymbol{B} = \sum_{i,I} A_{iI} B_{iI} = \mathrm{tr}(\boldsymbol{A}^T \boldsymbol{B}).$$

Objectivity and isotropy

Frame indifference

For every proper orthogonal spatial rotation $\boldsymbol{Q}$,

$$W(\boldsymbol{Q}\boldsymbol{F}_\chi) = W(\boldsymbol{F}_\chi).$$

An objective energy may be written in terms of $\boldsymbol{C} = \boldsymbol{F}_\chi^T \boldsymbol{F}_\chi$.

Material isotropy

For every proper orthogonal material rotation $\boldsymbol{R}$,

$$W(\boldsymbol{F}_\chi \boldsymbol{R}) = W(\boldsymbol{F}_\chi).$$

An objective isotropic energy may be written using the invariants of $\boldsymbol{C}$.

An anisotropic material can remain objective. Its energy also depends on material directions or structural tensors.

Compressible and incompressible response

Exact incompressibility

$$J = 1$$

is a constraint. A pressure-like Lagrange multiplier contributes

$$-p_s \, J \boldsymbol{F}_\chi^{-T}$$

to the first Piola stress.

Nearly incompressible model

$$W(\boldsymbol{F}_\chi) = W_{\text{iso}}(\bar{\boldsymbol{F}}_\chi) + U(J)$$

uses a volumetric energy to discourage numerical volume change.

$$\dot{J} = J \, (\nabla_x \cdot \boldsymbol{u}) \circ \chi.$$

At the continuum level, no slip and $\nabla_x \cdot \boldsymbol{u} = 0$ preserve the initial value $J(\boldsymbol{X}, 0)$. They give $J \equiv 1$ only when $J(\boldsymbol{X}, 0) = 1$. Discrete IBFE solutions can still develop local volume error.

Isochoric and volumetric deformation

In spatial dimension d, define

$$\boxed{\bar{\boldsymbol{F}}_\chi = J^{-1/d}\boldsymbol{F}_\chi}, \qquad \det \bar{\boldsymbol{F}}_\chi = 1.$$

Then

$$\bar{\boldsymbol{C}} = \bar{\boldsymbol{F}}_\chi{}^T \bar{\boldsymbol{F}}_\chi,$$

and the modified invariants of $\bar{\boldsymbol{C}}$ describe the isochoric response.

$$\bar{\boldsymbol{F}}_\chi = J^{-1/2}\boldsymbol{F}_\chi \quad \text{in 2D}, \qquad\qquad \bar{\boldsymbol{F}}_\chi = J^{-1/3}\boldsymbol{F}_\chi \quad \text{in 3D}.$$

Equivalently, in 3D the motion splits multiplicatively into a volumetric and a distortional part,

$$\boldsymbol{F}_\chi = J^{1/3}\bar{\boldsymbol{F}}_\chi, \qquad \bar{\boldsymbol{F}}_\chi = J^{-1/3}\boldsymbol{F}_\chi.$$

Building the distortional energy from modified invariants prevents the shear response from also penalizing volume change.

Griffith & Patankar (2020, §2.2.2) report that modified invariants together with volumetric stabilization give accuracy comparable to stabilized finite-element formulations on the Bonet et al. (2015) twisted-beam benchmark.

Volumetric stabilization

A common positive-J choice is

$$U(J) = \frac{\kappa_{\text{stab}}}{2}(\ln J)^2.$$

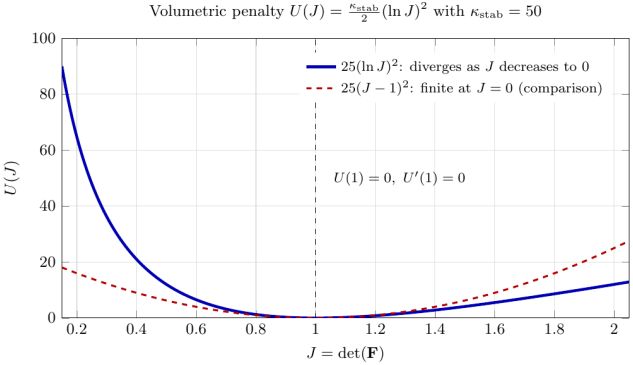
It satisfies

$$U(1) = 0, \qquad U'(1) = 0.$$

Unlike a quadratic $(J-1)^2$ penalty, $(\ln J)^2$ diverges as $J \to 0^+$: the barrier shown in the plot.

Its Piola-stress contribution is

$$\boldsymbol{P}_{\text{vol}} = U'(J) J \boldsymbol{F}_\chi^{-T} = \kappa_{\text{stab}} \ln J \, \boldsymbol{F}_\chi^{-T}.$$



In an incompressible IBFE model, κ_{stab} controls numerical volume error. It need not equal a physical bulk modulus.

Compressible neo-Hookean energy

One common three-dimensional model is

$$\boxed{W(\boldsymbol{F}_\chi) = \frac{\mu}{2}(I_1 - 3) - \mu \ln J + \frac{\lambda_{\mathrm{L}}}{2}(\ln J)^2},$$

where

$$I_1 = \operatorname{tr}(\boldsymbol{F}_\chi^T \boldsymbol{F}_\chi), \qquad J = \det \boldsymbol{F}_\chi > 0.$$

- μ is the infinitesimal shear modulus.
- λ_{L} is the first Lamé parameter.
- The infinitesimal bulk modulus is $K = \lambda_{\mathrm{L}} + 2\mu/3$.

This distinction keeps a material Lamé parameter separate from the optional stabilization coefficient κ_{stab}. In an (nearly) incompressible IBFE model, the $(\ln J)^2$ term of the neo-Hookean W plays exactly the role of κ_{stab}.

Neo-Hookean first Piola stress

For the energy on the previous slide,

$$\boxed{\boldsymbol{P} = \mu \left(\boldsymbol{F}_\chi - \boldsymbol{F}_\chi^{-T} \right) + \lambda_{\mathrm{L}} \ln J \, \boldsymbol{F}_\chi^{-T}}.$$

Reference state

At $\boldsymbol{F}_\chi = \boldsymbol{I}$,

$$J = 1, \qquad \boldsymbol{P} = \boldsymbol{0}.$$

Isochoric state

If $J = 1$,

$$\boldsymbol{P} = \mu(\boldsymbol{F}_\chi - \boldsymbol{F}_\chi^{-T}).$$

The final section derives this expression term by term.

Neo-Hookean stress conversions

Once $\boldsymbol{P}$ is known,

$$\boldsymbol{S} = \boldsymbol{F}_\chi^{-1}\boldsymbol{P}, \qquad \boldsymbol{\sigma} = J^{-1}\boldsymbol{P}\boldsymbol{F}_\chi^T.$$

For the compressible neo-Hookean model,

$$\boxed{\boldsymbol{\sigma} = \frac{\mu}{J}(\boldsymbol{B} - \boldsymbol{I}) + \frac{\lambda_\mathrm{L} \ln J}{J}\boldsymbol{I}}.$$

$\boldsymbol{\sigma}$ is symmetric because $\boldsymbol{B}$ and $\boldsymbol{I}$ are symmetric. The same constitutive response appears different because each stress uses a different configuration and area measure.

IBFE force pipeline

1. Evaluate $\boldsymbol{F}_{\chi h} = \nabla_{\boldsymbol{X}} \chi_h$, J_h, and $\boldsymbol{P}_h = \partial W / \partial \boldsymbol{F}_{\chi}$ at structural quadrature points.
2. Define the unified Lagrangian force by negative virtual work:

$$\int_{B_0} \boldsymbol{F}_h \cdot \boldsymbol{V}_h \, dV = - \int_{B_0} \boldsymbol{P}_h : \nabla_{\boldsymbol{X}} \boldsymbol{V}_h \, dV.$$

3. Spread $\boldsymbol{F}_h$ to the Eulerian grid with a regularized delta kernel.
4. Advance the incompressible fluid equations.
5. Restrict the Eulerian velocity back to the structure and update χ_h.

The negative sign makes internal elastic force oppose increasing stored energy. Prescribed body forces or boundary tractions add their own virtual-work terms.

Finite-element interpolation of motion and stress

With structural basis functions $\{\phi_\ell(\boldsymbol{X})\}$,

$$\chi_h(\boldsymbol{X}, t) = \sum_{\ell=1}^{M} \chi_\ell(t)\phi_\ell(\boldsymbol{X}),$$

$$\boldsymbol{F}_{\chi h}(\boldsymbol{X}, t) = \sum_{\ell=1}^{M} \chi_\ell(t) \otimes \nabla_{\boldsymbol{X}}\phi_\ell(\boldsymbol{X}).$$

- C^0 interpolation makes χ_h continuous.
- $\boldsymbol{F}_{\chi h}$ and $\boldsymbol{P}_h$ are generally discontinuous across element boundaries.
- Constitutive stress and interaction forces are evaluated at element quadrature points.
- Interaction quadrature may be refined independently of structural nodes and helps prevent leakage when adequately resolved.

Module 4 wrap-up

- Hyperelasticity defines $\boldsymbol{P} = \partial W / \partial \boldsymbol{F}_\chi$.
- The identity $\delta W = \boldsymbol{P} : \delta \boldsymbol{F}_\chi$ defines the matrix derivative.
- Objectivity and isotropy impose different constitutive requirements.
- Modified invariants and $U(J)$ separate shape and volume response.
- The weak structural force carries a negative virtual-work sign.
- FE quadrature connects piecewise constitutive fields to the Eulerian grid.

Next, one homogeneous deformation will pass through the entire chain from χ to $\boldsymbol{\sigma}$.

Worked example: uniaxial stretch (prescribed axial deformation)

Consider the three-dimensional motion

$$\chi(\boldsymbol{X}) = \begin{bmatrix} \alpha X_1 \\ X_2 \\ X_3 \end{bmatrix}, \qquad \alpha > 0.$$

- Material points stretch by the factor α along X_1.
- The transverse coordinates remain unchanged.
- There is no shear.

Throughout the course this example is called *uniaxial stretch (prescribed axial deformation)*; it is sometimes also called uniaxial strain. It is not automatically a uniaxial-stress test because the lateral stretches are fixed.

Axial deformation gradient and Jacobian

Differentiate each component of the motion:

$$\boldsymbol{F}_\chi = \nabla_X \chi = \begin{bmatrix} \frac{\partial \chi_1}{\partial X_1} & \frac{\partial \chi_1}{\partial X_2} & \frac{\partial \chi_1}{\partial X_3} \\ \frac{\partial \chi_2}{\partial X_1} & \frac{\partial \chi_2}{\partial X_2} & \frac{\partial \chi_2}{\partial X_3} \\ \frac{\partial \chi_3}{\partial X_1} & \frac{\partial \chi_3}{\partial X_2} & \frac{\partial \chi_3}{\partial X_3} \end{bmatrix} = \text{diag}(\alpha, 1, 1).$$

Therefore

$$\boxed{J = \det \boldsymbol{F}_\chi = \alpha}.$$

The prescribed deformation changes volume whenever $\alpha \neq 1$.

Axial Cauchy–Green tensor and Green strain

Because $\boldsymbol{F}_\chi$ is diagonal,

$$\boldsymbol{C} = \boldsymbol{F}_\chi^T \boldsymbol{F}_\chi = \text{diag}(\alpha^2, 1, 1).$$

Then

$$\boldsymbol{E} = \frac{1}{2}(\boldsymbol{C} - \boldsymbol{I}) = \text{diag}\left(\frac{\alpha^2 - 1}{2}, 0, 0\right).$$

$$E_{11} = \frac{\alpha^2 - 1}{2}. \qquad E_{22} = E_{33} = 0.$$

For $\alpha = 1 + \varepsilon$ with $|\varepsilon| \ll 1$, $E_{11} = \varepsilon + O(\varepsilon^2)$.

Axial-deformation invariants

For

$$\boldsymbol{C} = \mathrm{diag}(\alpha^2, 1, 1),$$

the first and third invariants are

$$I_1 = \mathrm{tr}\, \boldsymbol{C} = \alpha^2 + 2,$$

$$I_3 = \det \boldsymbol{C} = \alpha^2 = J^2.$$

The second invariant is

$$I_2 = 1 + 2\alpha^2.$$

I_3 directly records volume change. The raw invariants I_1 and I_2 reflect both the axial shape change and the accompanying volume change.

Constitutive choice for the axial example

Use the compressible neo-Hookean energy

$$W(\boldsymbol{F}_\chi) = \frac{\mu}{2}(I_1 - 3) - \mu \ln J + \frac{\lambda_\mathrm{L}}{2}(\ln J)^2.$$

For the prescribed axial deformation,

$$I_1 = \alpha^2 + 2, \qquad J = \alpha,$$

so

$$W(\alpha) = \frac{\mu}{2}(\alpha^2 - 1) - \mu \ln \alpha + \frac{\lambda_\mathrm{L}}{2}(\ln \alpha)^2.$$

The stress still follows by differentiating the full matrix-valued argument $\boldsymbol{F}_\chi$, not only the reduced scalar expression $W(\alpha)$.

First Piola stress for the axial deformation

Use

$$\boldsymbol{P} = \mu(\boldsymbol{F}_\chi - \boldsymbol{F}_\chi^{-T}) + \lambda_{\mathrm{L}} \ln J \, \boldsymbol{F}_\chi^{-T}.$$

Here

$$\boldsymbol{F}_\chi^{-T} = \mathrm{diag}(\alpha^{-1}, 1, 1), \qquad \ln J = \ln \alpha.$$

Therefore

$$\boxed{\boldsymbol{P} = \mathrm{diag}\left(\mu(\alpha - \alpha^{-1}) + \lambda_{\mathrm{L}} \frac{\ln \alpha}{\alpha}, \lambda_{\mathrm{L}} \ln \alpha, \lambda_{\mathrm{L}} \ln \alpha\right).}$$

Fixing both lateral stretches to one generally produces nonzero transverse stress.

Cauchy stress for the axial deformation

Use the Piola transformation:

$$\boldsymbol{\sigma} = \frac{1}{J}\boldsymbol{P}\boldsymbol{F}_{\chi}^{T}.$$

Since $J = \alpha$ and both tensors are diagonal,

$$\boldsymbol{\sigma} = \operatorname{diag}\left(P_{11}, \frac{P_{22}}{\alpha}, \frac{P_{33}}{\alpha}\right).$$

Thus

$$\sigma_{22} = \sigma_{33} = \frac{\lambda_{\mathrm{L}} \ln \alpha}{\alpha}.$$

A traction-free uniaxial-stress test must allow unknown lateral stretches and solve for them from $\sigma_{22} = \sigma_{33} = 0$.

Axial-example wrap-up

$$\chi(\boldsymbol{X}) \longrightarrow \boldsymbol{F}_\chi = \mathrm{diag}(\alpha, 1, 1) \longrightarrow J = \alpha$$

$$\longrightarrow \boldsymbol{C} = \mathrm{diag}(\alpha^2, 1, 1) \longrightarrow \boldsymbol{E} = \frac{1}{2}\,\mathrm{diag}(\alpha^2 - 1, 0, 0).$$

After choosing W,

$$W(\boldsymbol{F}_\chi) \longrightarrow \boldsymbol{P} = \frac{\partial W}{\partial \boldsymbol{F}_\chi} \longrightarrow \boldsymbol{\sigma} = J^{-1}\boldsymbol{P}\boldsymbol{F}_\chi^T.$$

Every arrow can be checked independently. Homogeneous deformations are useful unit tests for constitutive implementations.

Derivation goal

Start with

$$W(\boldsymbol{F}_\chi) = \frac{\mu}{2}(I_1 - 3) - \mu \ln J + \frac{\lambda_\mathrm{L}}{2}(\ln J)^2.$$

We will compute the first variation and match

$$\delta W = \boldsymbol{P} : \delta \boldsymbol{F}_\chi.$$

Two identities drive the calculation:

$$\delta I_1 = 2\boldsymbol{F}_\chi : \delta \boldsymbol{F}_\chi, \qquad \delta(\ln J) = \boldsymbol{F}_\chi^{-T} : \delta \boldsymbol{F}_\chi.$$

The next two slides derive these identities before the terms are combined.

Step 1: variation of I_1

Because

$$I_1 = \mathrm{tr}(\boldsymbol{F}_\chi^T \boldsymbol{F}_\chi) = \sum_{i,I} (F_\chi)_{iI}^2,$$

$$\delta I_1 = 2 \sum_{i,I} (F_\chi)_{iI}\, \delta (F_\chi)_{iI} = \boxed{2\boldsymbol{F}_\chi : \delta \boldsymbol{F}_\chi}.$$

Therefore

$$\delta \left[\frac{\mu}{2} (I_1 - 3) \right] = \mu \boldsymbol{F}_\chi : \delta \boldsymbol{F}_\chi.$$

This term contributes $\mu \boldsymbol{F}_\chi$ to $\boldsymbol{P}$.

Step 2: variation of $\ln J$

Jacobi's determinant identity gives

$$\delta J = J \boldsymbol{F}_{\chi}^{-T} : \delta \boldsymbol{F}_{\chi}.$$

The chain rule then gives

$$\boxed{\delta(\ln J) = \frac{\delta J}{J} = \boldsymbol{F}_{\chi}^{-T} : \delta \boldsymbol{F}_{\chi}}.$$

Consequently,

$$\delta(-\mu \ln J) = -\mu \boldsymbol{F}_{\chi}^{-T} : \delta \boldsymbol{F}_{\chi},$$

$$\delta\left[\frac{\lambda_{\mathrm{L}}}{2}(\ln J)^2\right] = \lambda_{\mathrm{L}} \ln J \, \boldsymbol{F}_{\chi}^{-T} : \delta \boldsymbol{F}_{\chi}.$$

Step 3: combine the stress contributions

Collect the coefficients of $\delta \boldsymbol{F}_\chi$:

$$\delta W = \left[\mu \boldsymbol{F}_\chi - \mu \boldsymbol{F}_\chi^{-T} + \lambda_{\mathrm{L}} \ln J \, \boldsymbol{F}_\chi^{-T} \right] : \delta \boldsymbol{F}_\chi .$$

Matching $\delta W = \boldsymbol{P} : \delta \boldsymbol{F}_\chi$ gives

$$\boxed{\boldsymbol{P} = \mu(\boldsymbol{F}_\chi - \boldsymbol{F}_\chi^{-T}) + \lambda_{\mathrm{L}} \ln J \, \boldsymbol{F}_\chi^{-T}} .$$

$\boldsymbol{F}_\chi = \boldsymbol{I} \quad \Rightarrow \quad \boldsymbol{P} = \boldsymbol{0}.$ $\qquad$ $J = 1 \quad \Rightarrow \quad \boldsymbol{P} = \mu(\boldsymbol{F}_\chi - \boldsymbol{F}_\chi^{-T}).$

Micro-check: small-strain limit

Let

$$\boldsymbol{F}_\chi = \boldsymbol{I} + \boldsymbol{H}, \qquad \|\boldsymbol{H}\| \ll 1, \qquad \boldsymbol{\varepsilon} = \operatorname{sym} \boldsymbol{H}.$$

To first order,

$$\boldsymbol{F}_\chi^{-T} \approx \boldsymbol{I} - \boldsymbol{H}^T, \qquad \ln J \approx \operatorname{tr} \boldsymbol{H}.$$

Substitution gives

$$\boldsymbol{P} \approx \mu(\boldsymbol{H} + \boldsymbol{H}^T) + \lambda_{\mathrm{L}} \operatorname{tr}(\boldsymbol{H})\boldsymbol{I} = \boxed{2\mu\boldsymbol{\varepsilon} + \lambda_{\mathrm{L}} \operatorname{tr}(\boldsymbol{\varepsilon})\boldsymbol{I}}.$$

At first order, $\boldsymbol{P}$ and $\boldsymbol{\sigma}$ coincide. The model reduces to isotropic linear elasticity.

Micro-check: axial-deformation plug-in

Use

$$\boldsymbol{F}_\chi = \mathrm{diag}(\alpha, 1, 1), \quad J = \alpha, \quad \boldsymbol{F}_\chi^{-T} = \mathrm{diag}(\alpha^{-1}, 1, 1).$$

Task

Insert these expressions into

$$\boldsymbol{P} = \mu(\boldsymbol{F}_\chi - \boldsymbol{F}_\chi^{-T}) + \lambda_{\mathrm{L}} \ln J \, \boldsymbol{F}_\chi^{-T}$$

and compute P_{11}, P_{22}, P_{33}.

$$P_{11} = \mu(\alpha - \alpha^{-1}) + \lambda_{\mathrm{L}} \frac{\ln \alpha}{\alpha}, \qquad P_{22} = P_{33} = \lambda_{\mathrm{L}} \ln \alpha.$$

These lateral stresses vanish only for $\alpha = 1$ under the prescribed lateral stretches.

References and further reading

Primary IB and IBFE references

- B. E. Griffith and X. Luo, *Hybrid finite difference/finite element immersed boundary method*, Int. J. Numer. Methods Biomed. Eng. 33 (2017), e2888. doi:10.1002/cnm.2888
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- C. S. Peskin, *The immersed boundary method*, Acta Numerica 11 (2002), 479–517. doi:10.1017/S0962492902000077

Continuum mechanics

- G. A. Holzapfel, *Nonlinear Solid Mechanics*, Wiley, 2000.
- J. Bonet and R. D. Wood, *Nonlinear Continuum Mechanics for Finite Element Analysis*, Cambridge, 2008.

Open resources

- Ajeet Kumar solid-mechanics notes
- NPTEL continuum-mechanics course
- IBAMR documentation and source